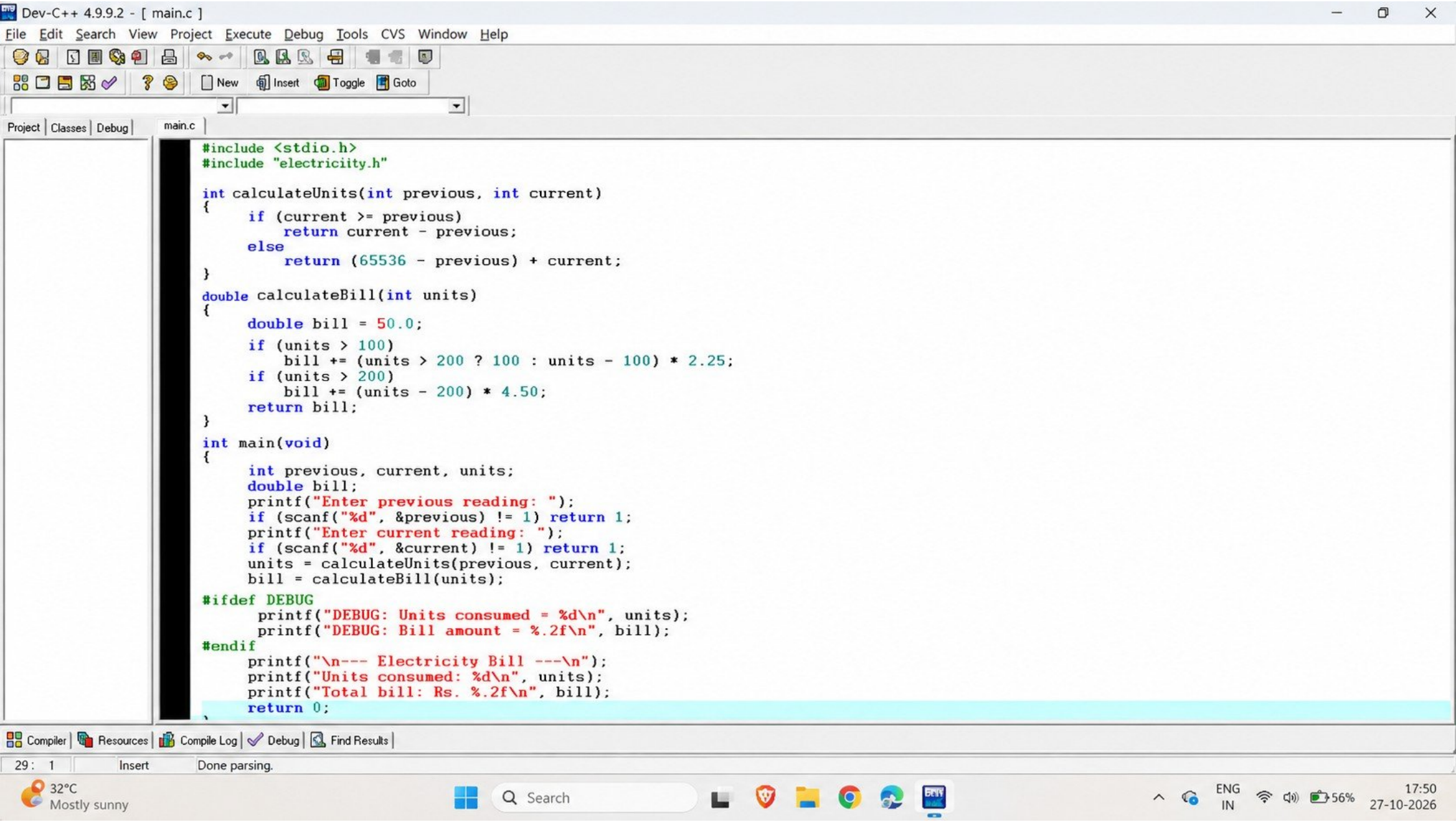
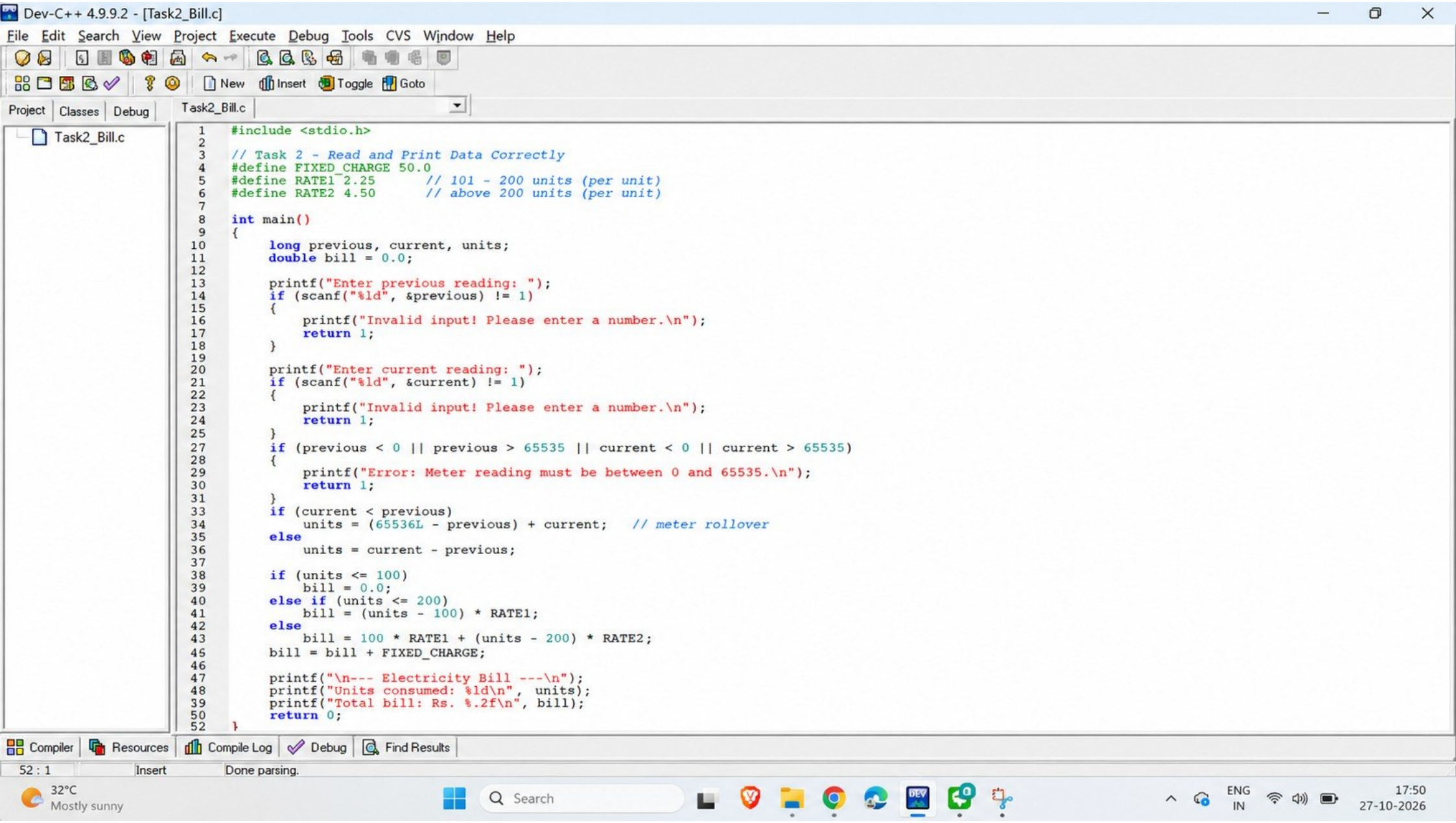
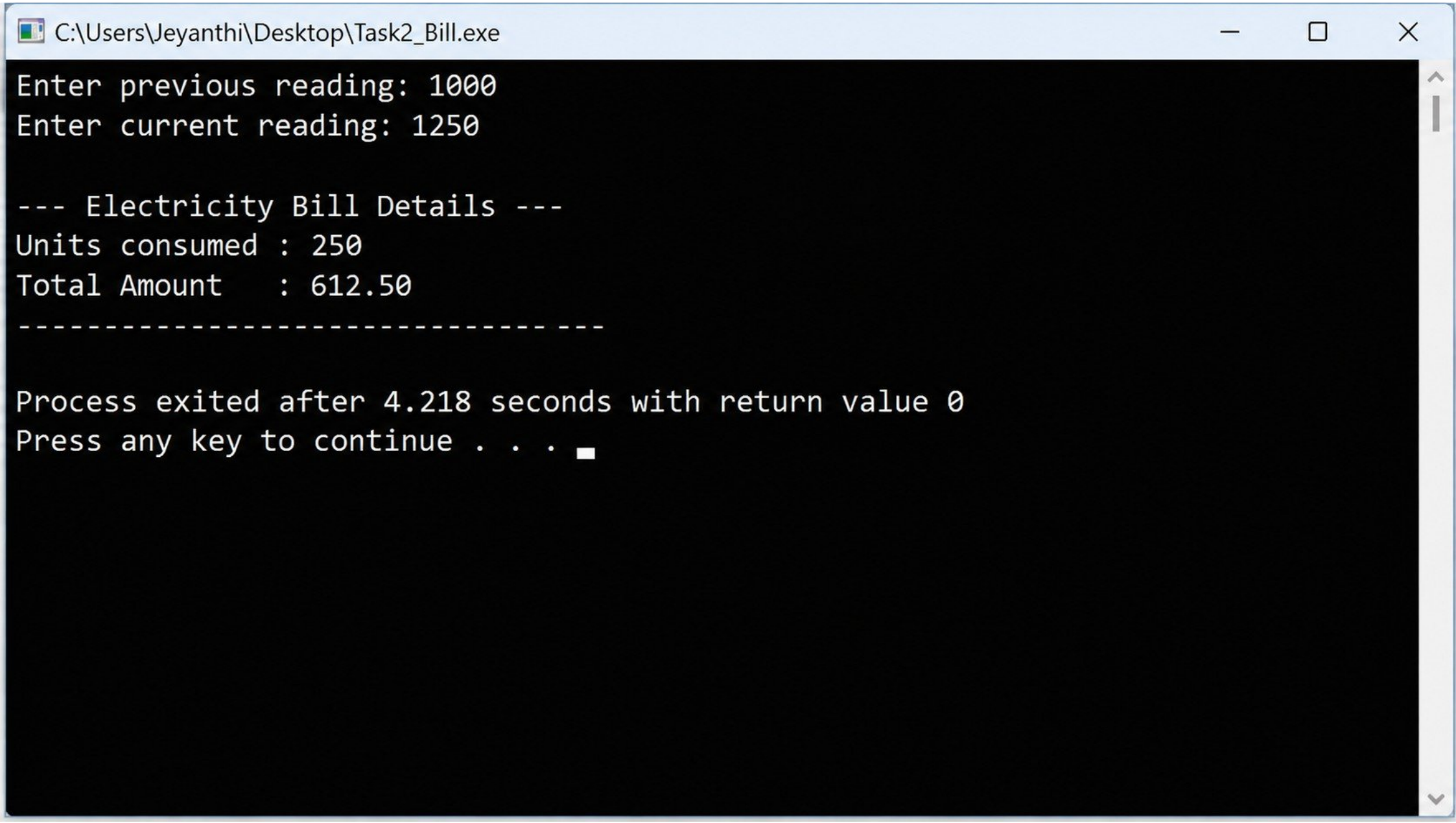

Dev-C++ 4.9.9.2 - [ main.c ]
File Edit Search View Project Execute Debug Tools CVS Window Help
New Insert Toggle Goto
Project Classes Debug
main.c
#include <stdio.h>
#include "electriciity.h"
int calculateUnits(int previous, int current)
{
if (current >= previous)
return current - previous;
else
return (65536 - previous) + current;
}
double calculateBill(int units)
{
double bill = 50.0;
if (units > 100)
bill += (units > 200 ? 100 : units - 100) * 2.25;
if (units > 200)
bill += (units - 200) * 4.50;
return bill;
}
int main(void)
{
int previous, current, units;
double bill;
printf("Enter previous reading: ");
if (scanf("%d", &previous) != 1) return 1;
printf("Enter current reading: ");
if (scanf("%d", &current) != 1) return 1;
units = calculateUnits(previous, current);
bill = calculateBill(units);
#ifdef DEBUG
printf("DEBUG: Units consumed = %d\n", units);
printf("DEBUG: Bill amount = %.2f\n", bill);
#endif
printf("\n--- Electricity Bill ---\n");
printf("Units consumed: %d\n", units);
printf("Total bill: Rs. %.2f\n", bill);
return 0;
Compiler Resources Compile Log Debug Find Results
29 : 1 Insert Done parsing.
32°C Mostly sunny
Search
ENG IN
56%
17:50 27-10-2026

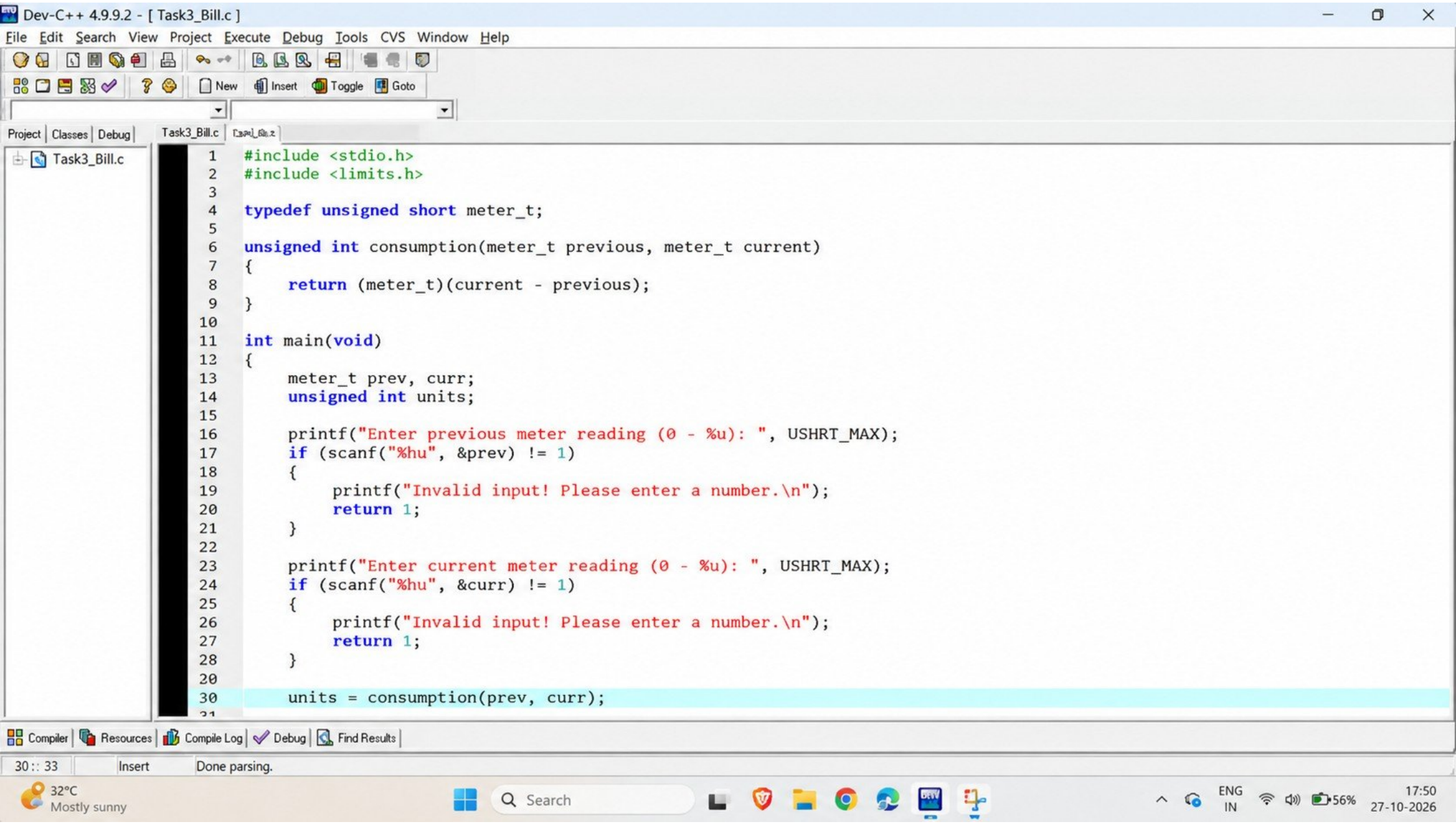
```
Enter previous reading: 1000
Enter current reading: 1250
DEBUG: Units consumed = 250
DEBUG: Bill amount = 612.50

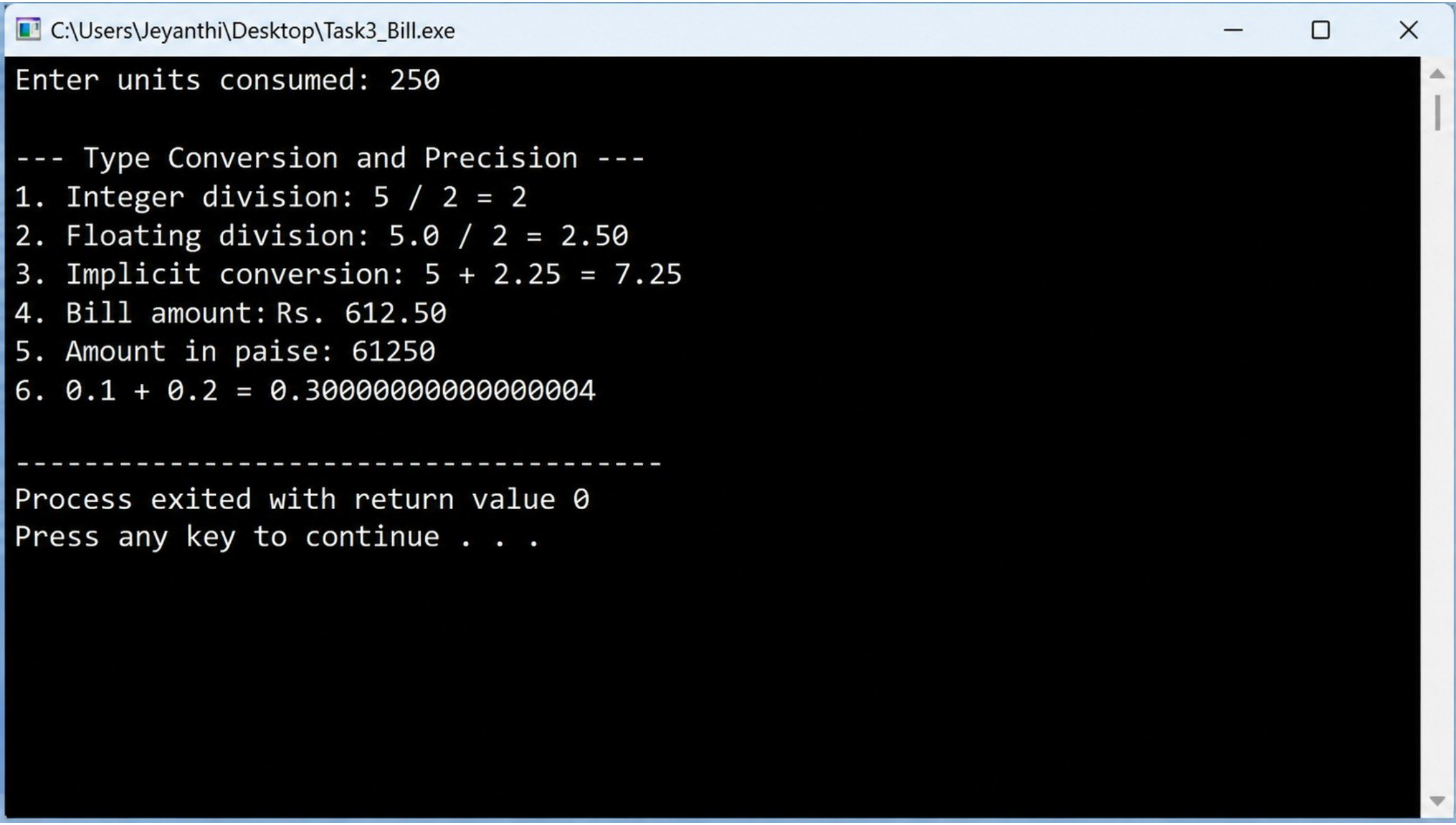
--- Electricity Bill ---
Units consumed: 250
Total bill: Rs. 612.50

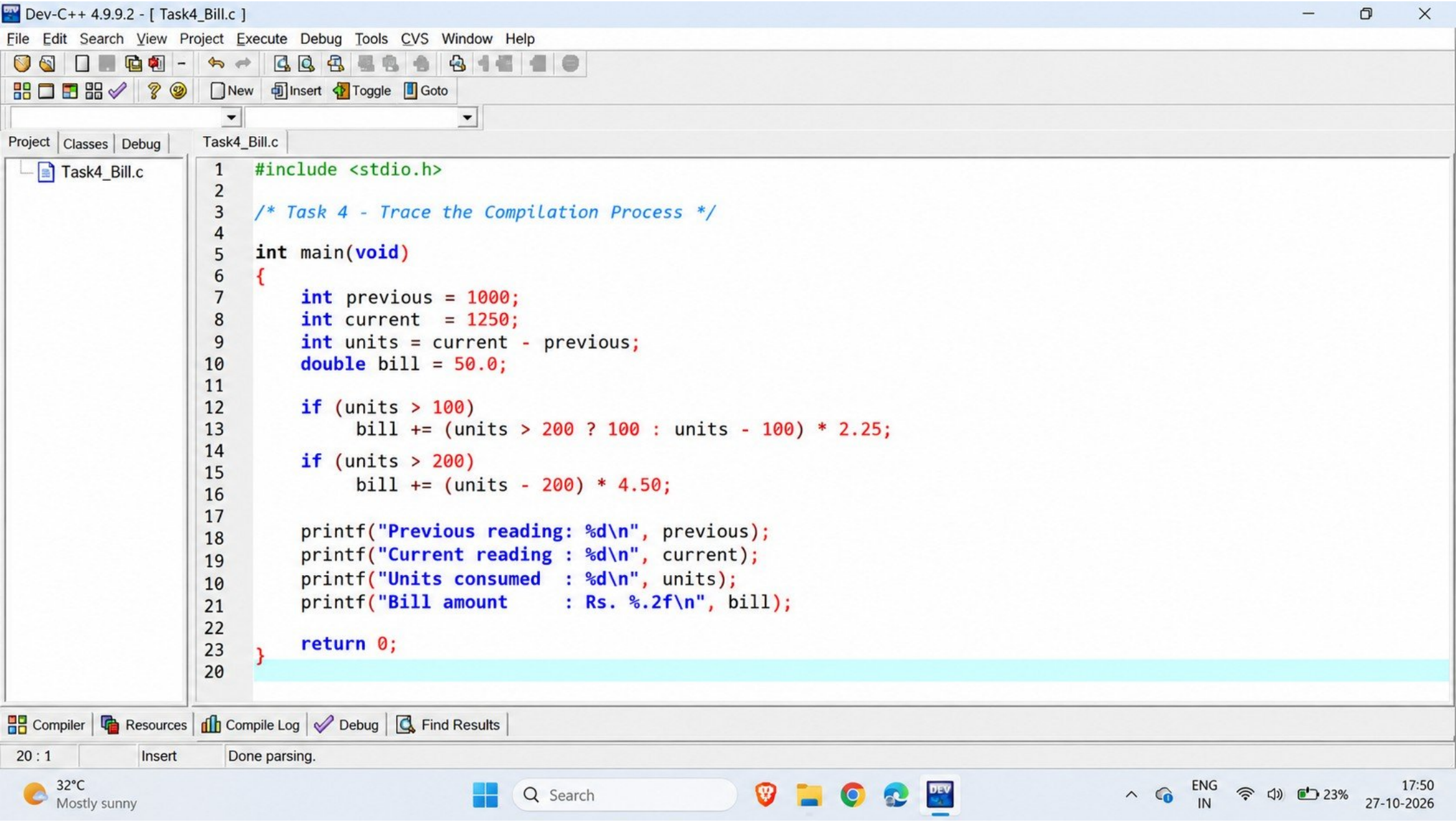
--------------------------------
Process exited after 7.321 seconds with return value 0
Press any key to continue . . .
```

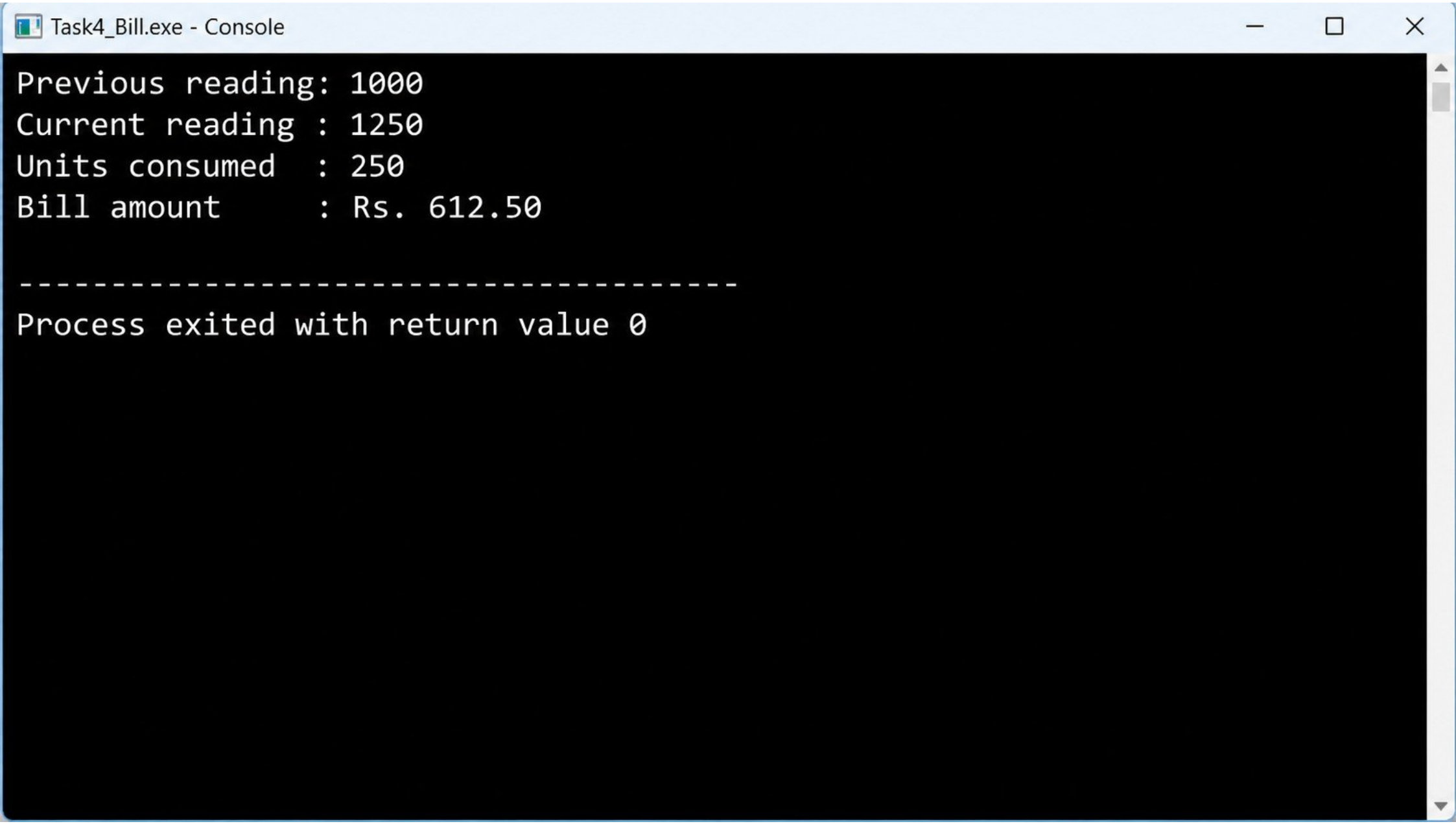

Dev-C++ 4.9.9.2 - [Task2_Bill.c]
File Edit Search View Project Execute Debug Tools CVS Window Help
New Insert Toggle Goto
Project Classes Debug
Task2_Bill.c
Task2_Bill.c
#include <stdio.h>
// Task 2 - Read and Print Data Correctly
#define FIXED_CHARGE 50.0
#define RATE1 2.25 // 101 - 200 units (per unit)
#define RATE2 4.50 // above 200 units (per unit)
int main()
{
long previous, current, units;
double bill = 0.0;
printf("Enter previous reading: ");
if (scanf("%ld", &previous) != 1)
{
printf("Invalid input! Please enter a number.\n");
return 1;
}
printf("Enter current reading: ");
if (scanf("%ld", &current) != 1)
{
printf("Invalid input! Please enter a number.\n");
return 1;
}
if (previous < 0 || previous > 65535 || current < 0 || current > 65535)
{
printf("Error: Meter reading must be between 0 and 65535.\n");
return 1;
}
if (current < previous)
units = (65536L - previous) + current; // meter rollover
else
units = current - previous;
if (units <= 100)
bill = 0.0;
else if (units <= 200)
bill = (units - 100) * RATE1;
else
bill = 100 * RATE1 + (units - 200) * RATE2;
bill = bill + FIXED_CHARGE;
printf("\n--- Electricity Bill ---\n");
printf("Units consumed: %ld\n", units);
printf("Total bill: Rs. %.2f\n", bill);
return 0;
}
Compiler Resources Compile Log Debug Find Results
52 : 1 Insert Done parsing.
32°C Mostly sunny
Search
ENG IN
17:50 27-10-2026


C:\Users\Jeyanthi\Desktop\Task2_Bill.exe
Enter previous reading: 1000
Enter current reading: 1250
--- Electricity Bill Details ---
Units consumed : 250
Total Amount : 612.50
Process exited after 4.218 seconds with return value 0
Press any key to continue . . .


Dev-C++ 4.9.9.2 - [ Task3_Bill.c ]
File Edit Search View Project Execute Debug Tools CVS Window Help
New Insert Toggle Goto
Project Classes Debug
Task3_Bill.c
Task3_Bill.c
#include <stdio.h>
#include <limits.h>
typedef unsigned short meter_t;
unsigned int consumption(meter_t previous, meter_t current)
{
return (meter_t)(current - previous);
}
int main(void)
{
meter_t prev, curr;
unsigned int units;
printf("Enter previous meter reading (0 - %u): ", USHRT_MAX);
if (scanf("%hu", &prev) != 1)
{
printf("Invalid input! Please enter a number.\n");
return 1;
}
printf("Enter current meter reading (0 - %u): ", USHRT_MAX);
if (scanf("%hu", &curr) != 1)
{
printf("Invalid input! Please enter a number.\n");
return 1;
}
units = consumption(prev, curr);
Compiler Resources Compile Log Debug Find Results
30 :: 33 Insert Done parsing.
32°C Mostly sunny
Search
ENG IN
56%
17:50 27-10-2026


C:\Users\Jeyanthi\Desktop\Task3_Bill.exe
Enter units consumed: 250
--- Type Conversion and Precision ---
1. Integer division: 5 / 2 = 2
2. Floating division: 5.0 / 2 = 2.50
3. Implicit conversion: 5 + 2.25 = 7.25
4. Bill amount: Rs. 612.50
5. Amount in paise: 61250
6. 0.1 + 0.2 = 0.30000000000000004
Process exited with return value 0
Press any key to continue . . .


Dev-C++ 4.9.9.2 - [ Task4_Bill.c ]
File Edit Search View Project Execute Debug Tools CVS Window Help
New Insert Toggle Goto
Project Classes Debug
Task4_Bill.c
Task4_Bill.c
#include <stdio.h>
/* Task 4 - Trace the Compilation Process */
int main(void)
{
int previous = 1000;
int current = 1250;
int units = current - previous;
double bill = 50.0;
if (units > 100)
bill += (units > 200 ? 100 : units - 100) * 2.25;
if (units > 200)
bill += (units - 200) * 4.50;
printf("Previous reading: %d\n", previous);
printf("Current reading : %d\n", current);
printf("Units consumed : %d\n", units);
printf("Bill amount : Rs. %.2f\n", bill);
return 0;
}
Compiler Resources Compile Log Debug Find Results
20 : 1 Insert Done parsing.
32°C
Mostly sunny
Search
ENG
IN
23%
17:50
27-10-2026


Task4_Bill.exe - Console
Previous reading: 1000
Current reading : 1250
Units consumed  : 250
Bill amount     : Rs. 612.50
---
Process exited with return value 0